

Microcirculation activator

Hijuven[®] 200 Soft Capsules

(dl- α -Tocopheryl nicotinate preparation)

1. PRODUCT NAME

Hijuven 200 Soft Capsules

2. PRODUCT DESCRIPTION AND COMPOSITION

2.1 product description

Brand name	Dosage form and identification code	Appearance			Description
Hijuven 200 Soft Capsules	Soft Capsules	E200			Capsules Orange Contents
	E200	Long diameter (mm)	Short diameter (mm)	Weight (mg)	Yellowish white, viscous suspension or semisolid
		14.7	7.2	465	

2.2 composition

Each orange soft capsule contains 200 mg of dl- α -tocopheryl nicotinate

3. INDICATION

Hyperlipidemia
Peripheral circulatory disturbances accompanying arteriosclerosis obliterans

4. DOSAGE AND ADMINISTRATION

- The usual adult dosage for oral use is 3 capsules daily in three divided doses.
- The dosage may be adjusted depending on the patient's age and symptoms.
- The usual adult dosage of dl- α -tocopheryl nicotinate for oral use is 300 - 600 mg daily in three divided doses.

5. CONTRAINDICATIONS

Hijuven is contraindicated in patients with hypersensitivity to any of the active or inactive ingredients in Hijuven.

6. WARNINGS AND PRECAUTIONS

Caution in handing over drug
For drugs that are dispensed in a press-through package (PTP), instruct the patient to remove the drug from the package prior to use. [It has been reported that, if the PTP sheet is swallowed, the sharp corners of the sheet may puncture the esophageal mucosa, causing perforation and resulting in serious complications such as mediastinitis.]

7. INTERACTION WITH OTHER MEDICAMENTS

No data to date.

8. FERTILITY, PREGNANCY AND LACTATION

No clinical data on fertility and excretion in human milk have been reported. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy. There is a very limited amount of data from the use in pregnant women. As a precautionary measure, it is preferable to avoid the use in such cases.

9. ADVERSE REACTIONS

Adverse reactions were reported in 105 of 5,621 patients (1.87 %). (Sum of the clinical trials)

	5% > $\geq$ 0.1%	<0.1%	Incidence unknown
Gastrointestinal	Anorexia, stomach discomfort, stomachache, nausea, diarrhea and constipation		
Hypersensitivity ^{note)}		Rash	
Hepatic		Liver Dysfunction (Elevation of AST (GOT) and ALT (GPT), etc.)	
Others		Feeling of warmth, flushing	Facial edema, edema

Note) In the event of such symptoms, Hijuven should be discontinued or other appropriate measures taken.

10. OVERDOSE AND TREATMENT

There is insufficient information on overdose.

11. PHARMACOLOGICAL PROPERTIES

11.1 Pharmacodynamics

Improvement of lipid metabolism

- In aged or cholesterol-loaded rats, dl- α -tocopheryl nicotinate decreases serum total cholesterol concentrations by increasing the turnover rate of cholesterol, which is considered to be due to enhanced catabolism/excretion of cholesterol. dl- α -Tocopheryl nicotinate also decreases lipid peroxide and triglyceride concentrations.
- dl- α -Tocopheryl nicotinate decreases total serum cholesterol and triglyceride levels in humans, while it increases the serum levels of high-density lipoprotein (HDL) cholesterol in lipoprotein metabolism.

Enhancement of microcirculation

- In an experiment with unanesthetized rabbits, dl- α -tocopheryl nicotinate has been demonstrated to maintain vasomotion and increase peripheral blood flow in the ear conchae by directly acting on the blood vessel walls, not via the nervous system.
- In a clinical trial, dl- α -tocopheryl nicotinate is more significantly effective in alleviating peripheral circulatory disturbances than coadministration of vitamin E and nicotinic acid.

Strengthening of blood vessels

dl- α -Tocopheryl nicotinate mitigates capillary hyperpermeability in humans and decreases the number of purpura spots.

Inhibition of platelet aggregation

- dl- α -Tocopheryl nicotinate inhibits the aggregation induced by epinephrine, arachidonic acid, collagen and ADP in stimulated human platelets.
- dl- α -Tocopheryl nicotinate is more potent than tocopherol and tocopherol acetate in inhibiting the aggregation induced by arachidonic acid and collagen in human platelet-rich plasma.

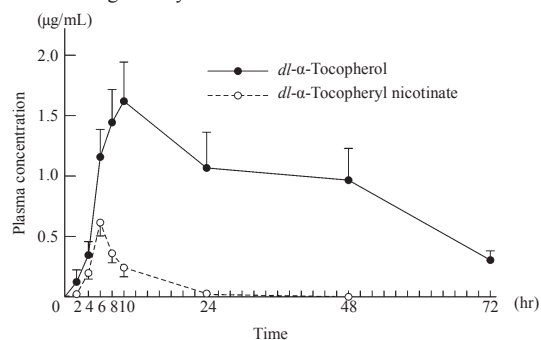
Restoration of arterial oxygen partial pressure

dl- α -Tocopheryl nicotinate restores the decreased partial pressure of oxygen in human arterial blood.

11.2 Pharmacokinetics

Blood concentrations

dl- α -Tocopheryl nicotinate was administered orally to 12 healthy adult male volunteers at a single dose of 600 mg^{note} after a meal, and plasma concentrations of dl- α -tocopheryl nicotinate (the unchanged drug) and tocopherol were determined. The plasma peak concentration (C_{max} =0.615 μ g/mL) of the unchanged drug was reached 6 hr after administration, and thereafter declined rapidly with an elimination half-life of 4.3 hr. The peak plasma tocopherol concentration (except endogenous tocopherol) (C_{max} =1.62 μ g/mL) was reached 10 hr after administration and declined gradually with an elimination half-life of 38.5 hr.



Plasma unchanged drug and α -tocopherol concentration after administration of tocopherol nicotinate

(Mean $\pm$ S.E., n=12)

Effect of meal

When dl- α -tocopheryl nicotinate was administered orally to 4 healthy adult male volunteers at a single dose of 600 mg^{note}, the peak plasma concentration was 32 times higher and the AUC was 29 times larger after a meal than during fasting. The absorption of dl- α -tocopheryl nicotinate is significantly influenced by food intake.

Note: A single dose of 600 mg is unapproved.

12. CLINICAL STUDIES

Clinical efficacy

Hyperlipidemia

In open labeled clinical trials, it has been shown that statistically significant reduction for the patients of high total cholesterol before administrations, and shown that significant increase for the patients of low HDL-cholesterol before administrations. And statistically significant reduction in serum lipid peroxides was observed in patients with hyperlipidemia.

Peripheral circulatory disturbances

In double-blind and open labeled clinical trials, dl- α -tocopheryl nicotinate has been demonstrated to be useful in patients with peripheral circulatory disturbances such as arteriosclerosis obliterans.

13. PHARMACEUTICAL PARTICULARS

13.1 List of excipients

L-aspartic acid, FD&C Yellow No.6 (Sunset Yellow - FCF), carnauba wax, glycerol esters of fatty acids, titanium oxide, gelatin, D-sorbitol solution, medium chain fatty acid triglyceride, concentrated glycerin, ethyl parahydroxybenzoate and propyl parahydroxybenzoate

13.2 Storage conditions

Store below 30°C.

13.3 Packaging

Boxes of 100 capsules (10 capsules $\times$ 10) in press-through packages.

14. MANUFACTURER

Shibakawa Plant of Fuji Capsule Co., Ltd.
789 Kamiyuno, Fujinomiya-shi, Shizuoka, Japan

15. PRODUCT REGISTRATION HOLDER

Eisai (Malaysia) Sdn. Bhd.
Unit 701D, Level 7, Tower D, Uptown 5
No. 5, Jalan SS21/39, Damansara Uptown, 47400 Petaling Jaya
Selangor, Malaysia

16. PRODUCT OWNER

Eisai Co., Ltd.
4-6-10 Koishikawa, Bunkyo-ku,
Tokyo, Japan



17. DATE OF REVISION OF PI

March 2022